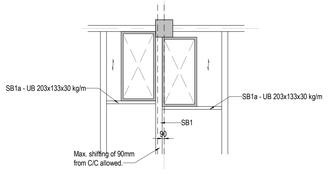
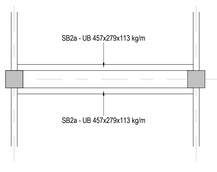
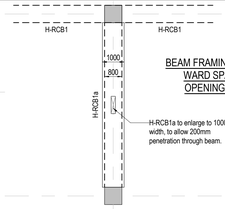


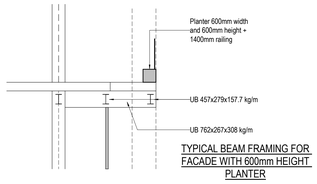
TYPICAL BEAM FRAMING ADJUSTMENT FOR FLOOR OPENING 1



TYPICAL BEAM FRAMING ADJUSTMENT FOR FLOOR OPENING 2



BEAM FRAMING ADJUSTMENT AT WARD SPACE FOR BEAM OPENING/PENETRATION



TYPICAL BEAM FRAMING FOR FACADE WITH 800mm HEIGHT PLANTER

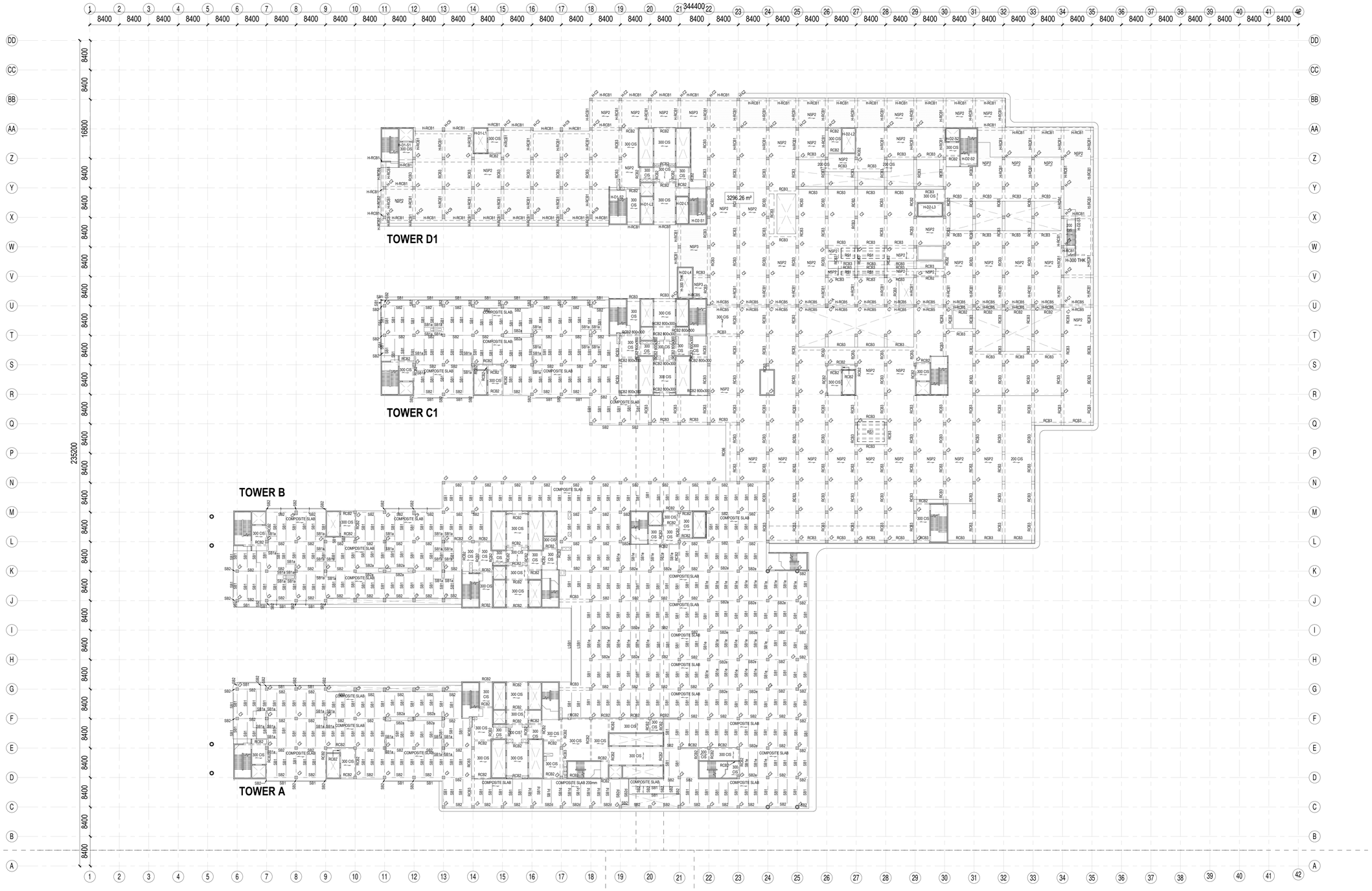
NOTES: (UNLESS OTHERWISE STATED):

1. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH RELEVANT ARCHITECTURAL AND M&E DRAWINGS.
2. ALL DIMENSION, SETTING OUT, DROPS, LEVEL & OPENING FOR SERVICES TO BE CHECKED AGAINST RELEVANT ARCHITECTURAL AND M&E DRAWINGS.
3. THIS DRAWING PROVIDE THE NOTIONAL STRUCTURAL SYSTEM AND CONCEPTS TO BE FURTHER DEVELOPED IN SCHEMATIC DESIGN - STAGE 2, THE CONTRACTOR SHALL BE RESPONSIBLE FOR ENSURING THAT THE FINAL DESIGN MEETS ALL REQUIREMENTS SET OUT IN THE LATEST ARCHITECTURAL, M&E AND TECHNICAL SPECIFICATIONS.
4. ALL CONCRETE GRADE SHALL REFER TO TGCH-TD-S-100-RD-01 DRAWING.
5. ALL REINFORCEMENT STEEL BAR SHALL BE GRADE 500.
6. ALL BEAMS SHALL BE 800(D)x800(W).
7. ALL NSP2 & NSP3 SHALL BE 130 THK + TOPPING 120 THK, ALL NSP5 TO BE 150 THK + TOPPING 200 THK.
8. ALL CIS SLAB SHALL BE 300 THK.
9. ALL COMPOSITE SLAB SHALL BE 140 THK WITH BONDEK II (1.0 THK) DECK OR EQUIVALENT MATERIAL.
10. ALL ELEMENTS MARKING WITH H-PREFIX SHALL BE HARDENING STRUCTURE AND SHALL BE CAST-IN SITU STRUCTURE.
11. ALL RS1 & WS1 SHALL REFER TO TGCH-TD-S-100-SD-06 DRAWING.
12. ALL WALLS SHALL BE 250mm THK.
13. ALL STEEL MATERIALS SHALL BE IN COMPLIANCE WITH THE PROVISIONS OF BC1: 2012.
14. ALL STEEL GRADE SHALL BE MIN. S355.
15. CLASS 42 ELECTRODE SHALL BE ADOPTED FOR WELDING OF GRADE S355 STEEL.

MARK	SIZE	REMARKS
SB1	UB 457x191x105.7	1 SHEAR STUD (19mm DIA.) PER @ 200mm c/c, Response Factor <2
SB1a	UB 203x133x30	1 SHEAR STUD (19mm DIA.) PER @ 200mm c/c, steel beam for opening
SB1b	RHS 500x100x12	1 SHEAR STUD (19mm DIA.) PER @ 200mm c/c, steel beam for façade support with slab
SB1c	RHS 500x100x4	steel beam for façade support without slab
SB1d	UB 457x279x128	1 SHEAR STUD (19mm DIA.) PER @ 200mm c/c, steel beam for landscape
SB1e	UB 457x191x158	1 SHEAR STUD (19mm DIA.) PER @ 200mm c/c, Response Factor <1
SB2	UB 457x279x193.5	2 SHEAR STUD (19mm DIA.) PER @ 200mm c/c, Response Factor <2
SB2a	UB 457x279x113	steel beam for opening
SB2b	UB 457x279x213	steel beam for façade support with slab
SB2c	UB 457x152x52.3	steel beam for façade support without slab
SB2d	UB 457x279x235	2 SHEAR STUD (19mm DIA.) PER @ 200mm c/c, steel beam for landscape
SB2e	UB 457x279x213	2 SHEAR STUD (19mm DIA.) PER @ 200mm c/c, Response Factor <1
SB3	UB 457x191x067	2 SHEAR STUD (19mm DIA.) PER @ 200mm c/c
SB3b	UB 457x279x128	2 SHEAR STUD (19mm DIA.) PER @ 200mm c/c, steel beam for façade support with slab
SB4	UB 457x279x193.5	2 SHEAR STUD (19mm DIA.) PER @ 200mm c/c
SB4b	UB 457x279x213	steel beam for façade support with slab
LSB1	UB 762x381x284.2	-
LSB2	UB 762x381x284.2	-
LSB3	UB 914x419x416.7	-
LSB4	UB 1016x402x542	steel beam for landscape

LEGEND:

THESE HATCHED AREAS ARE SLABS WITH PREFIX "#H." TO BE DESIGNED TO ALLOW FOR ACCIDENTAL LOAD CASE DUE TO THE COLLAPSE OF SINGLE STOREY SLAB AT STOREY ABOVE



4TH STOREY OVERALL PLAN

1 : 400

Sheet of

BUILDING AND CONSTRUCTION AUTHORITY

APPROVED UNDER SECTION 5/ SECTION 5A OF THE BUILDING CONTROL ACT (CHAPTER 29)

Project Ref No :

for Commissioner of Building Control

TENDER DRAWING

PE'S CERTIFICATION

No.	Date	Amendments

EMPLOYER
Ministry of Health Singapore
16 College Road
Singapore 169854
Tel: +(65) 6325 9220

DEVELOPMENT AGENT
MOH Holdings Pte Ltd
1 North Buona Vista Link
#09-01 Elementum
Singapore 139691
Tel: +(65) 6822 0980

OPERATION AGENT
National University Hospital System
NUHS Tower Block
1E Kent Ridge Road
Singapore 119228
Tel: +(65) 6908 2222

PROJECT DELIVERY CONSULTANT
Asia Infrastructure Solutions Pte Ltd
230 Victoria Street
#12-01 Bugis Junction Towers
Singapore 188024
Tel: +(65) 6222 3888

SECURITY & BLAST CONSULTANT
Prostruct Consulting
SJ Campus
36 Cleantech Loop, #01-31
Singapore 636741
Tel: +(65) 6232 1360

ARCHITECT
SAA Architects Pte Ltd
SJ Campus
36 Cleantech Loop, #03-22
Singapore 636740
Tel: +(65) 6232 1338

MEDICAL PLANNING
SJ HealthCare Pte Ltd
SJ Campus
36 Cleantech Loop, #01-31
Singapore 636741

MECHANICAL & ELECTRICAL ENGINEER
Surbana Jurong Consultants Pte Ltd
SJ Campus
38 Cleantech Loop, #01-31
Singapore 636741
Tel: +(65) 6248 1270

C&S ENGINEER:
KTP Consultants Pte Ltd
Member of the Surbana Jurong Group
Surbana Jurong Campus
38 CleanTech Loop, Singapore 636741
Tel: +65 6248 1288 | E-mail: info@ktpworld.com
Website: www.ktpworld.com

PROJECT:

PROPOSED DEVELOPMENT OF 7 BLOCKS OF 9-STOREY HOSPITAL COMPRISING OF ACUTE HOSPITAL, SPECIALIST OUTPATIENT CLINIC, COMMUNITY HOSPITAL AND 3 BASEMENT LEVEL WITH TWO LEVELS OF CARPARKING ON LOT & MK NO: MK10-05493V(PT), MK10-05495T(PT), MK10-05512X(PT), MK10-00167T(PT) AT TENGAH GARDEN AVENUE/TENGAH CENTRAL (TENGAH PLANNING AREA)

CIVIL AND STRUCTURAL

Drawing Title:

4TH STOREY OVERALL PLAN

APPROVED BY:	DESIGNED:	DRAWN BY:	CHECKED BY:
TY	YS/EW	KIN	TOK

SCALE @ A0:	DRAWING NO.	REVISION
As indicated	TGCH-TD-S-200-L4-00	0
DATE: DEC 2025		